MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 11** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If a sector of a circle of radius 21 cm has central angle 60°, then the area of the corresponding major sector is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 1155 cm^2 (B) 231 cm^2 (C) 1386 cm^2 (D) 924 cm^2

Q2. If the ratio of the areas of two concentric circles is 1 : 4 and their boundary ring has width 7 cm, then the radius of the inner circle is: [1]

(A) 7 cm (B) 14 cm (C) 3.5 cm (D) 21 cm

Q3. The perimeter of a sector of a circle of radius 10.5 cm with central angle 60° is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 32 cm (B) 11 cm (C) 21 cm (D) 42 cm

Q4. In a circle of radius r, the area of the triangle formed by two radii with central angle 120° is given by: [1]

(A) $\frac{\sqrt{3}}{4}r^2$ (B) $\frac{1}{2}r^2$ (C) $\frac{\sqrt{3}}{2}r^2$ (D) $\frac{1}{4}r^2$

Q5. If the perimeter of a circle is decreased by 50%, then its area is decreased by: [1]

(A) 25% (B) 50% (C) 75% (D) 100%

Q6. **Assertion (A):** If the circumference of a circle is 176 cm, then its radius is 28 cm.
Reason (R): The circumference C of a circle of radius r is given by $C = 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If the area of a circle inscribed in a square is 154 cm^2, find the area of the square. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

Q8. A car wiper blade of length 25 cm sweeps through an angle of 115°. Calculate the area cleaned in one sweep of this blade. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A chord of a circle of radius 10.5 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$ [3]

Q10. In a circle of radius 14 cm, a sector has central angle 90°. A semicircle is drawn on the chord of this sector as diameter. Find the area of the shaded crescent region. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. An Olympic archery target consists of five concentric scoring areas: Gold (d = 21 cm), Red, Blue, Black, and White, where each outer band is 10.5 cm wide.

(a) Calculate the radius of each concentric boundary.

(b) Calculate the area of each of the five individual scoring regions. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(c) Prove that the areas of the regions are in the arithmetic progression ratio 1 : 3 : 5 : 7 : 9.

[5]

Q12. In a square $ABCD$ of side 14 cm, four semicircles are drawn with each side as diameter, intersecting to form a four-petaled central flower.

(a) Label the four unshaded corner regions I, II, III, IV and derive the area of these unshaded regions.

(b) Calculate the total area of the four shaded petals. $\left(\text{Use } \pi = \frac{22}{7}\right)$

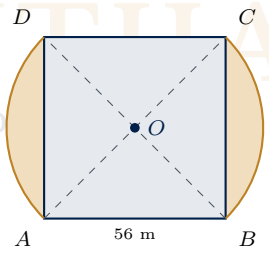
(c) Find the perimeter of the four shaded petals.

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Botanical Garden Semicircular Flower Beds on Lawn

A municipal botanical park features a central square green lawn $ABCD$ of side length 56 m. On two opposite sides, semicircular / circular sector flower beds are constructed, each centered at the diagonal intersection point O of the lawn.



(i) Find the distance from the lawn centre O to any corner vertex. [1]

(ii) Calculate the area of the central square green lawn. [1]

(iii) Calculate the combined total area of the lawn and both floral beds, OR calculate the area of the two floral beds alone. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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